Security 02 | EAP-TLS Wireless Policy

Overview

In this lab, we will be working within the `WiFi` configuration section of CV-CUE. Create an SSID (WPA2 802.1X) with your `ATD-##-EAP` as the name (where ## is a 2 digit character between 01-12 that was assigned to your lab/Pod).

What you'll learn:

- Configure a WPA2 802.1X EAP-TLS SSID in CV-CUE
- Create AGNI networks and segments for wireless EAP-TLS policy
- Use RadSec for secure RADIUS communication
- Enforce certificate-based authentication for wireless clients
- Monitor and verify client sessions in AGNI

Open Launchpad | **Login Help** | **Lab Topology**

Create an EAP-TLS SSID

The `Configure` section of CV-CUE is composed of multiple parts, including Wi-Fi, Alerts, WIPS, etc. In this lab, we are focused on the `WiFi` section.

Other configuration sections

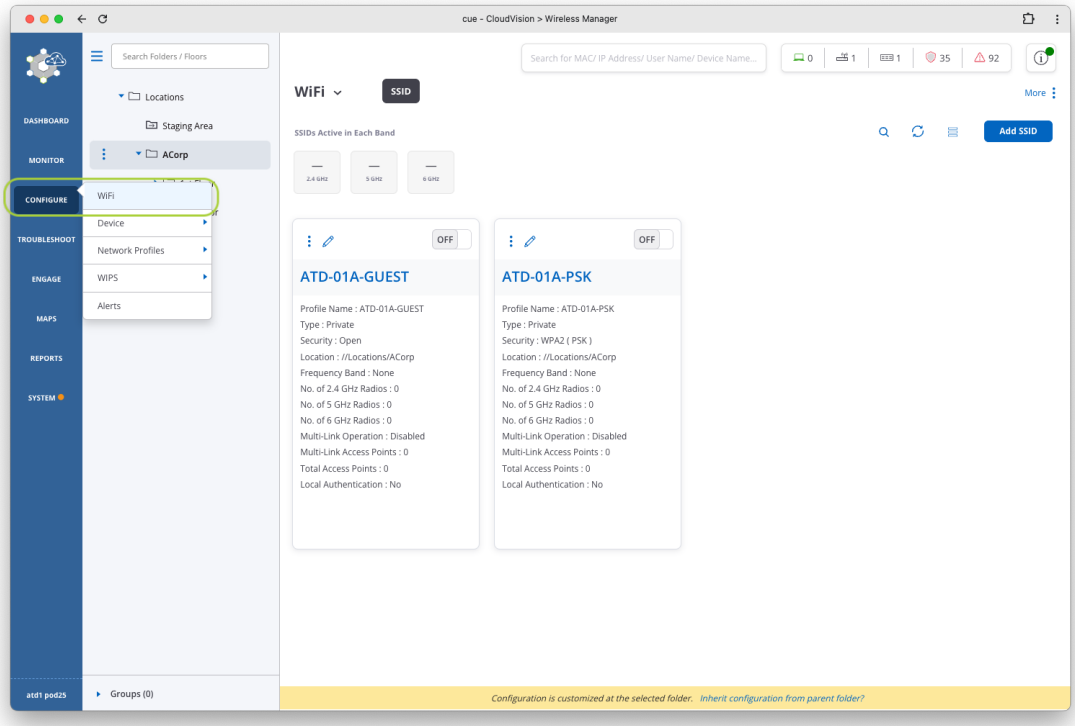
- **Alerts:** Where syslog and other alert related settings are configured
- **WIPS:** Where the policies are configured for the WIPS sensor.

1. Let's go through the steps to create a new SSID

Step 1

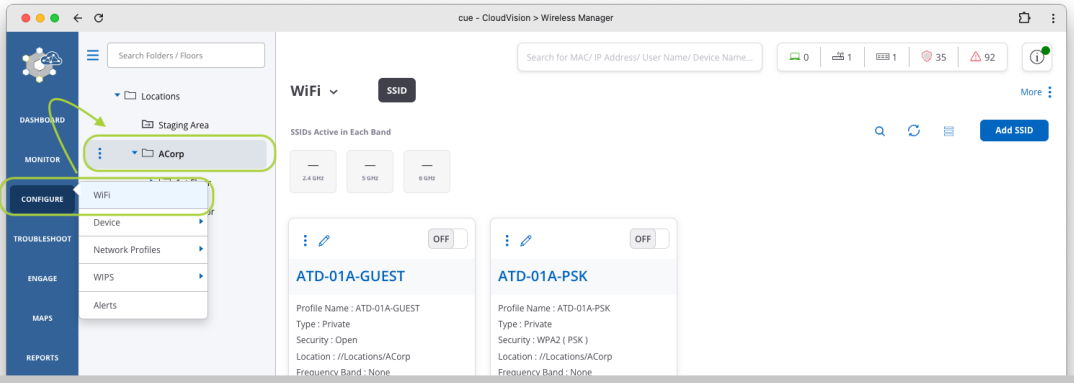
Hover your cursor over the `Configure` menu option on the left side of the screen, then select

`WiFi`



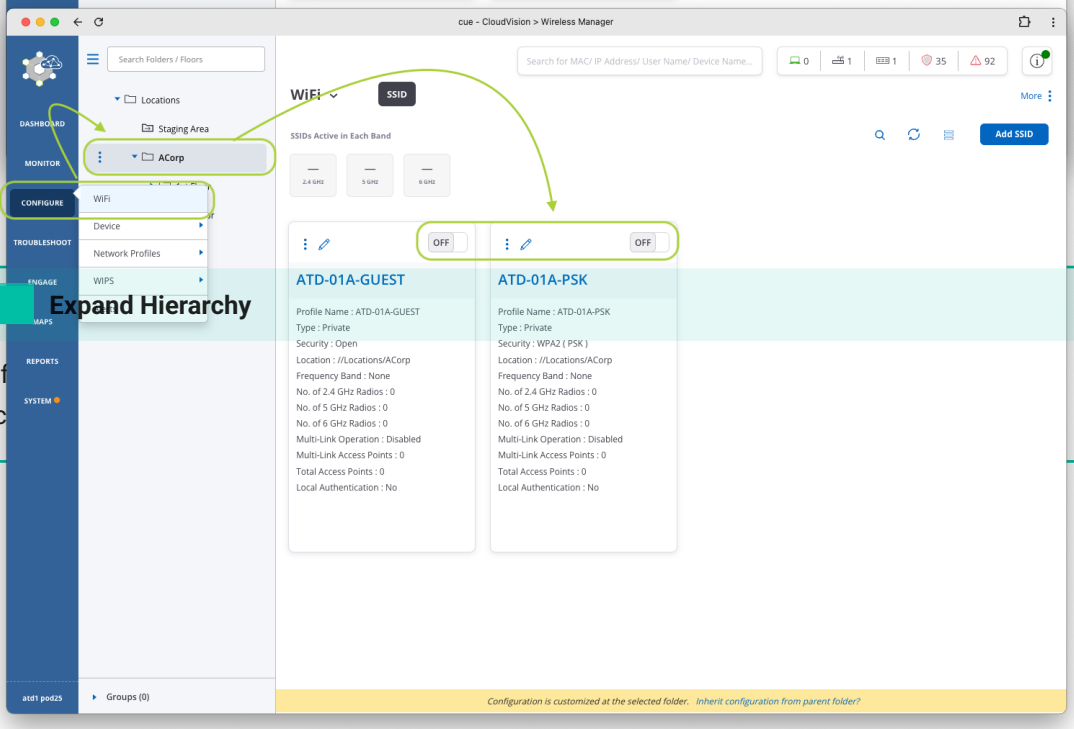
Step 2

At the top of the screen, you will see where you are in the location hierarchy. Select your respective `Corp` (`ACorp` or `BCorp`).



Step 3

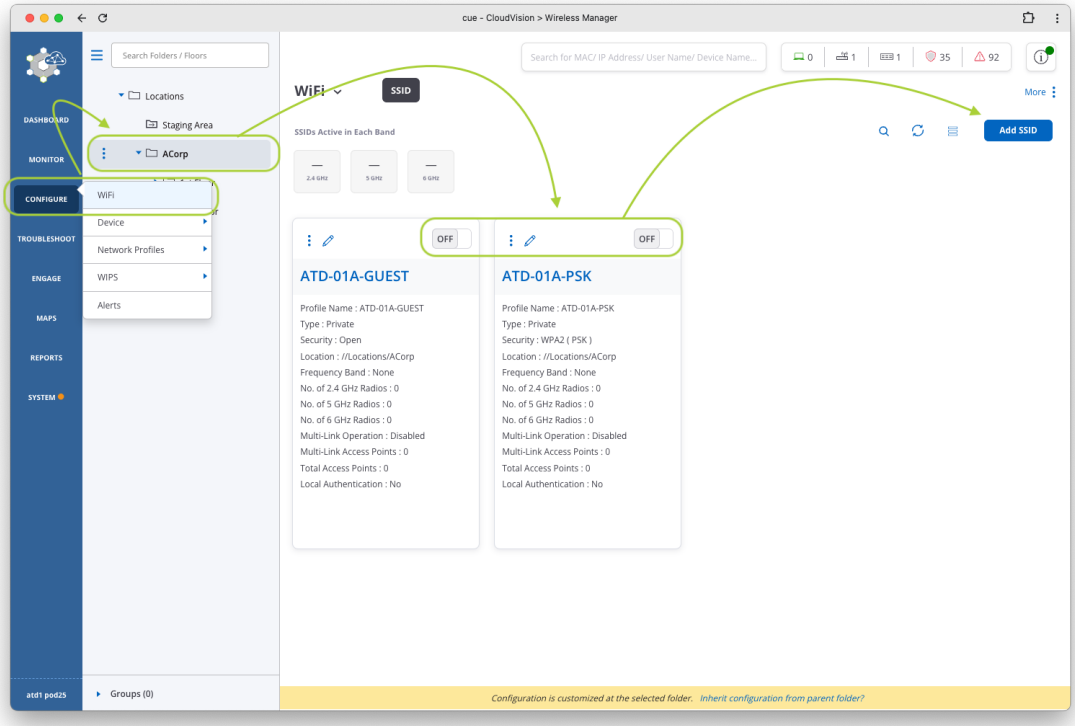
Let's disable the previous SSIDs from the prior labs by toggling the `On/Off` button



Expand Hierarchy

Step 4

Now select the `Add SSID` button on the right-hand side of the screen.

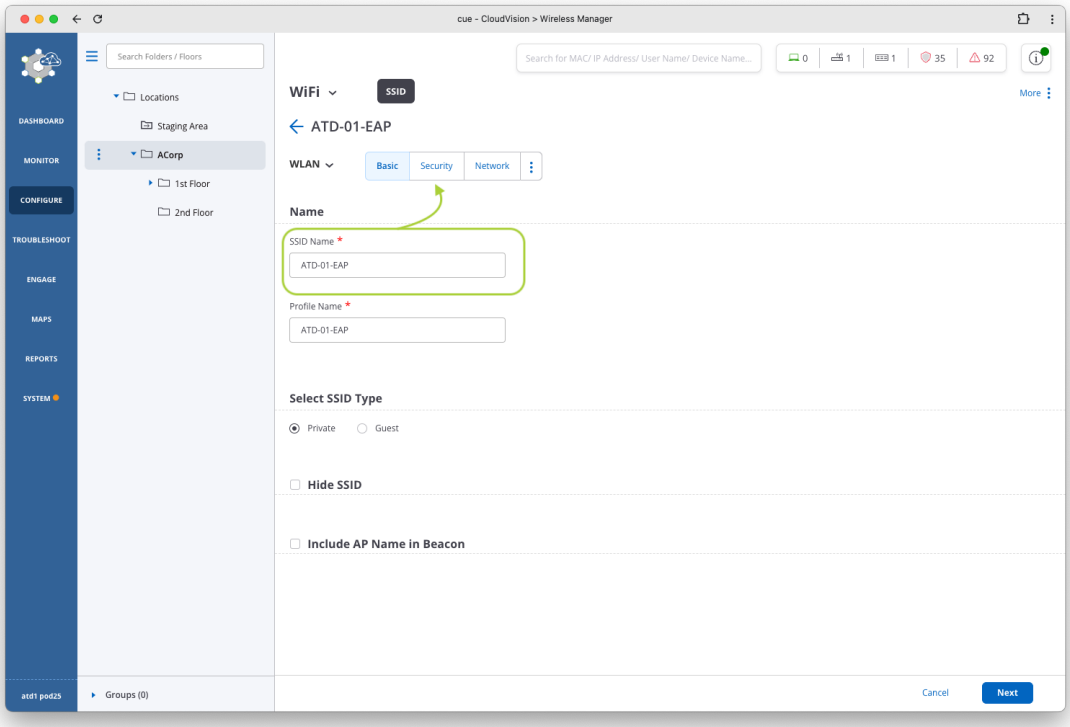


2. Once on the “SSID” page, configuration sub-category menu options will appear across the top of the page related to Wi-Fi (the defaults are `Basic`, `Security`, and `Network`). You can select these sub-category names to change configuration items related to that area of the configuration.
3. To make additional categories visible, select the 3 dots next to `Network` and you can see the other categories that are available to configure (`Analytics`, `Captive Portal`, etc.).
4. In the `Basic` sub-category option, name the SSID using the settings below. The `Profile Name` is used to describe the SSID and should have been auto-filled for you.

Settings

Student	Name
Student 1	`ATD-##A-EAP`
Student 2	`ATD-##B-EAP`

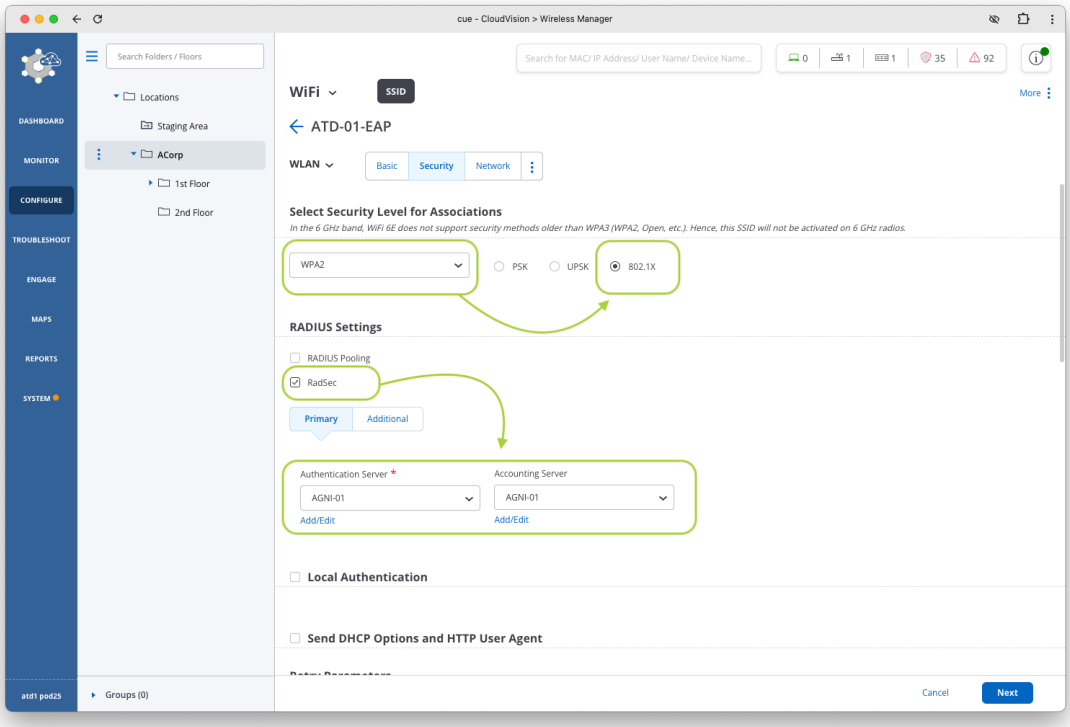
where ## is a 2 digit character between 01-12 that was assigned to your lab/Pod



5. Since this is our corporate SSID, leave the `Select SSID Type` set to `Private`
6. Select `Next` at the bottom.
7. In the `Security` sub-category, select `WPA2` and change the association type to `802.1X`.

Settings

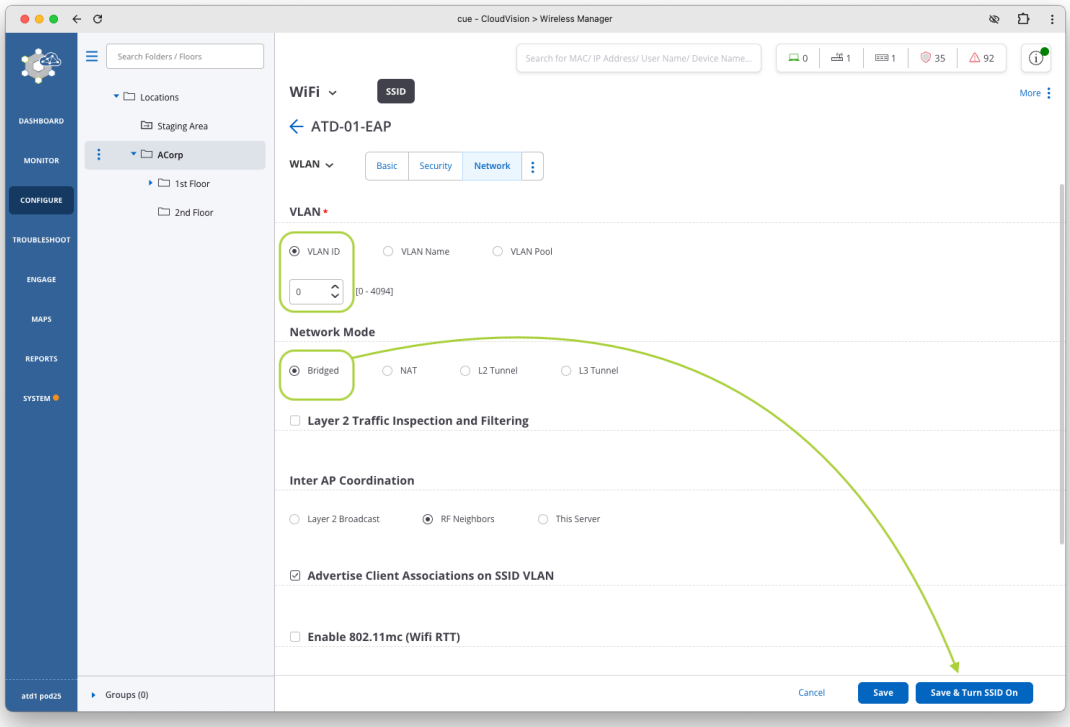
Field	Value
Security Method	WPA2 / 802.1X
Radius Settings	Select `RadSec`
Authentication Server	`AGNI-##`
Accounting Server	`AGNI-##`



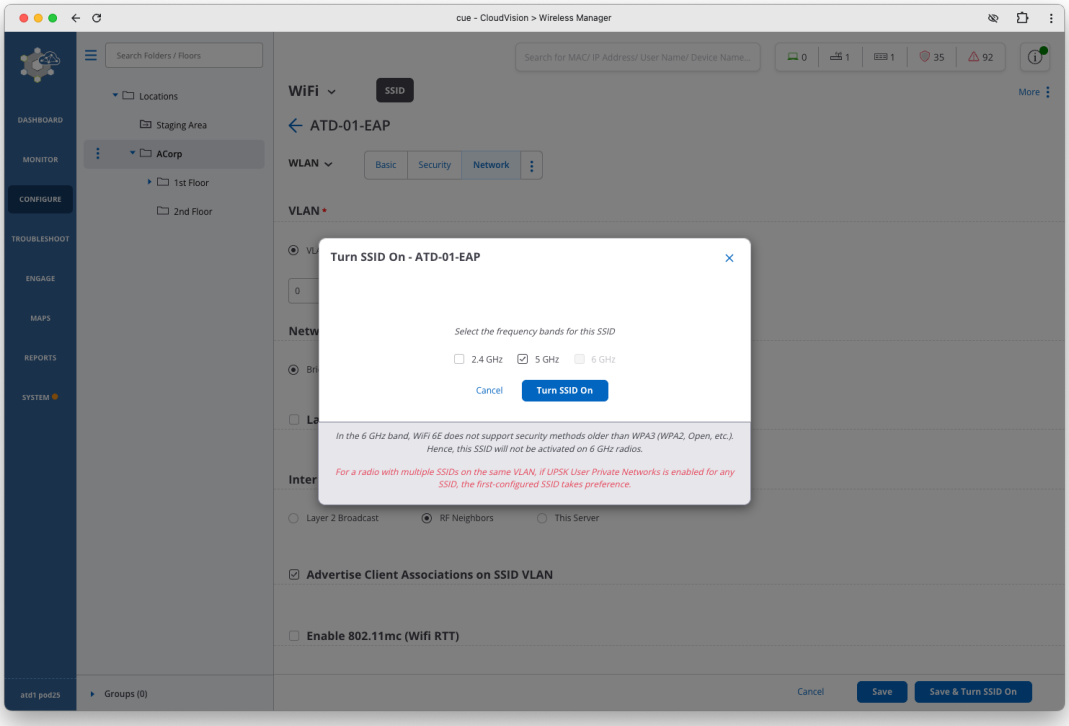
8. Select `Next` at the bottom of the screen.
9. In the `Network` configuration sub-category, we'll leave the `VLAN ID` set to `0`, which means it will use the native VLAN. If the switchport the AP is attached to is trunked, you could change this setting to whichever VLAN you want the traffic mapped to. The rest of the settings can be left at the default values.

Alternative Settings

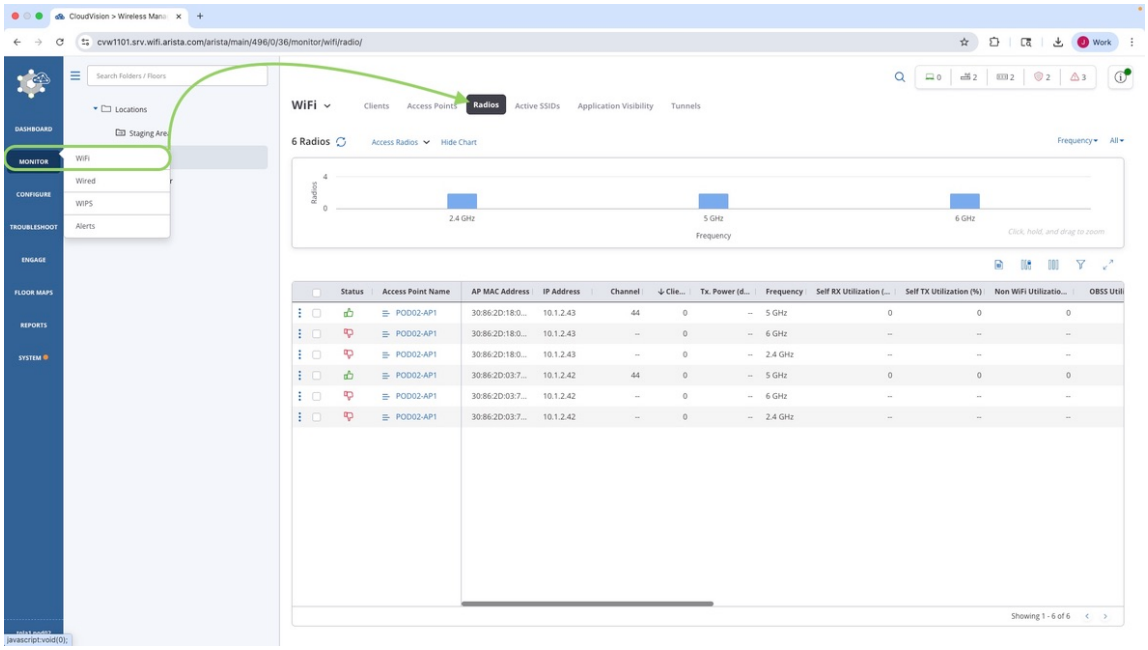
Instead of `Bridged` You could use `NAT` (often done for Guest) or `L2 Tunnel` / `L3 Tunnel`, as we completed in the wireless lab.



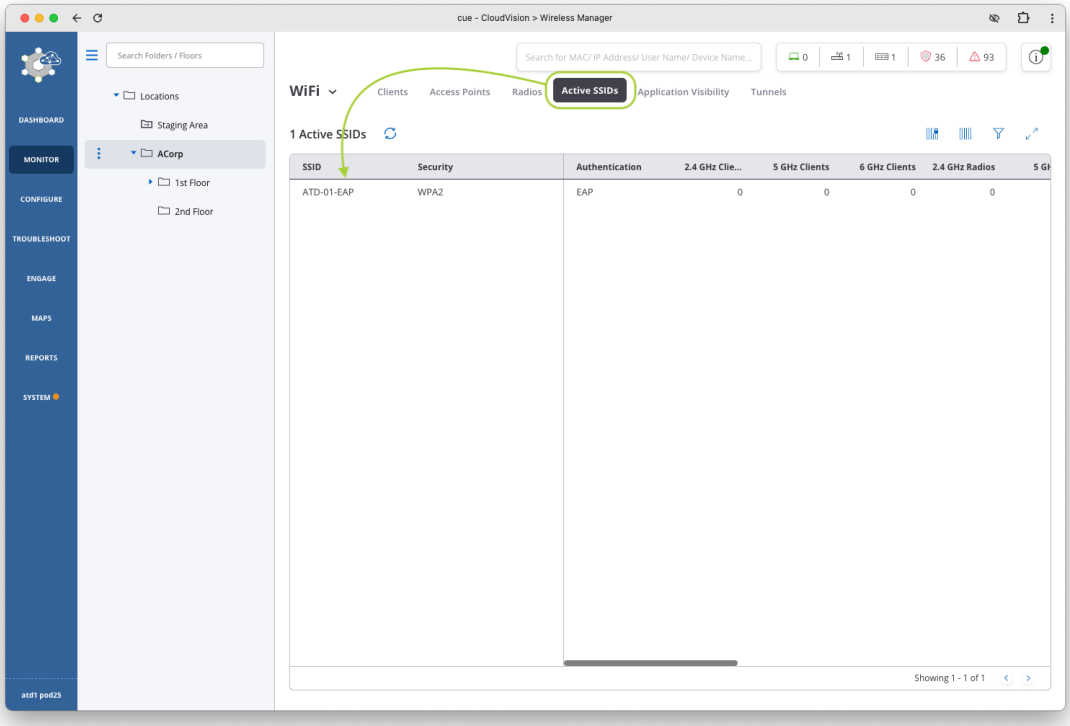
10. Select the `Save & Turn SSID On` button at the bottom of the page.
11. Only select the `5 GHz` option on the next screen (deselect the 2.4 GHz box if it's checked), then select `Turn SSID On`.



12. After you turn on the SSID, hover your cursor over `Monitor` in the left-hand side menu, and then select `WiFi`.
13. Now, in the menu options at the top of the page, look at the `Radios` menu option. Is the 5 GHz radio 👍 (Up) and 2.4 GHz radio 👎 (down)? It may take a minute or two for the radio to become active.



14. Check the `Active SSIDs` menu at the top of the screen. Is your SSID listed?



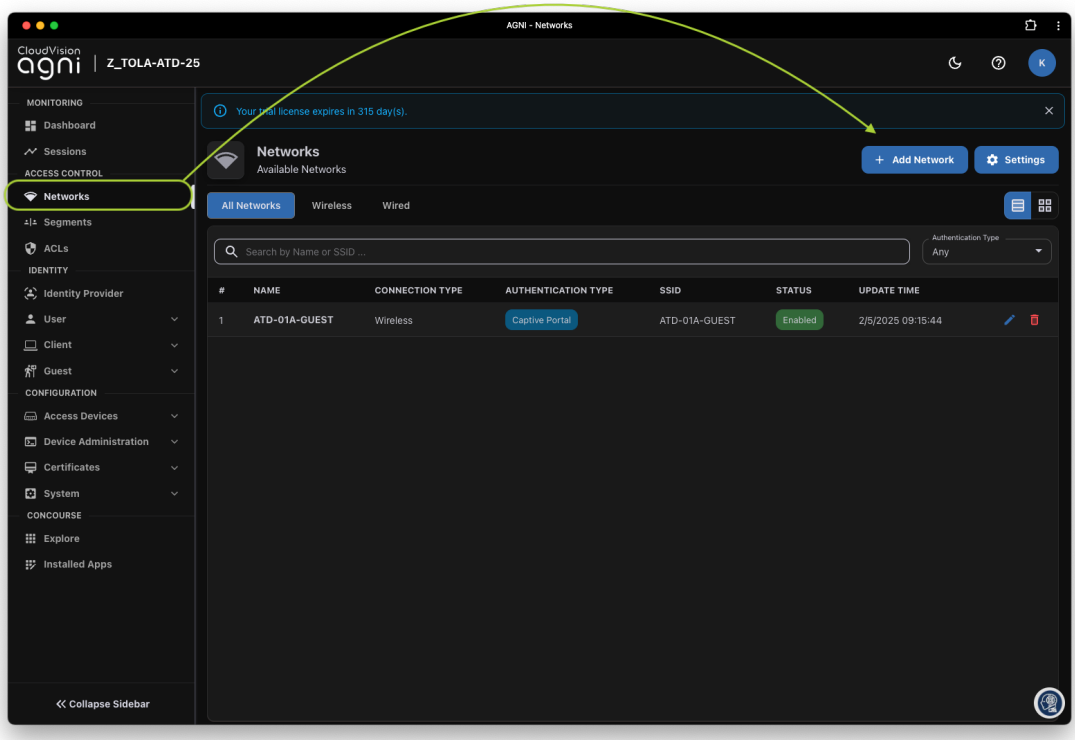
15. Now that we have a 802.1X backed SSID, let's go to AGNI to configure the policy.

CloudVision AGNI Access

Log in to **AGNI** from the Launchpad if you're not already there.

Create AGNI Networks & Segments for the EAP-TLS Wireless Policy

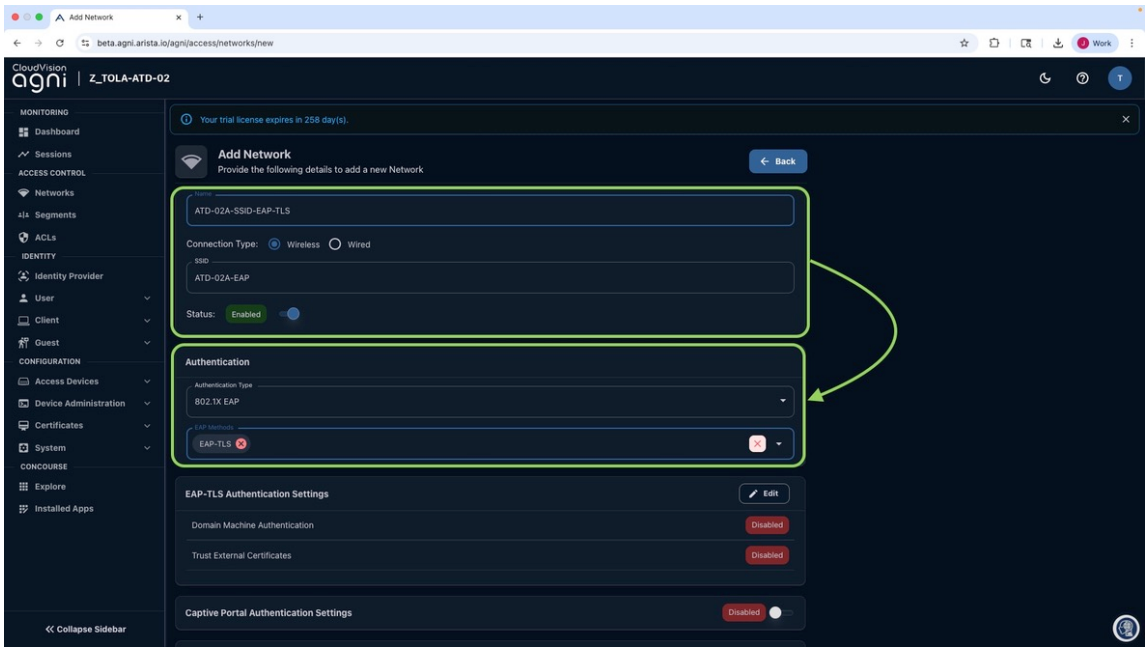
1. Select `Networks` and then select `+ Add Network`



2. Configure the network with the following settings

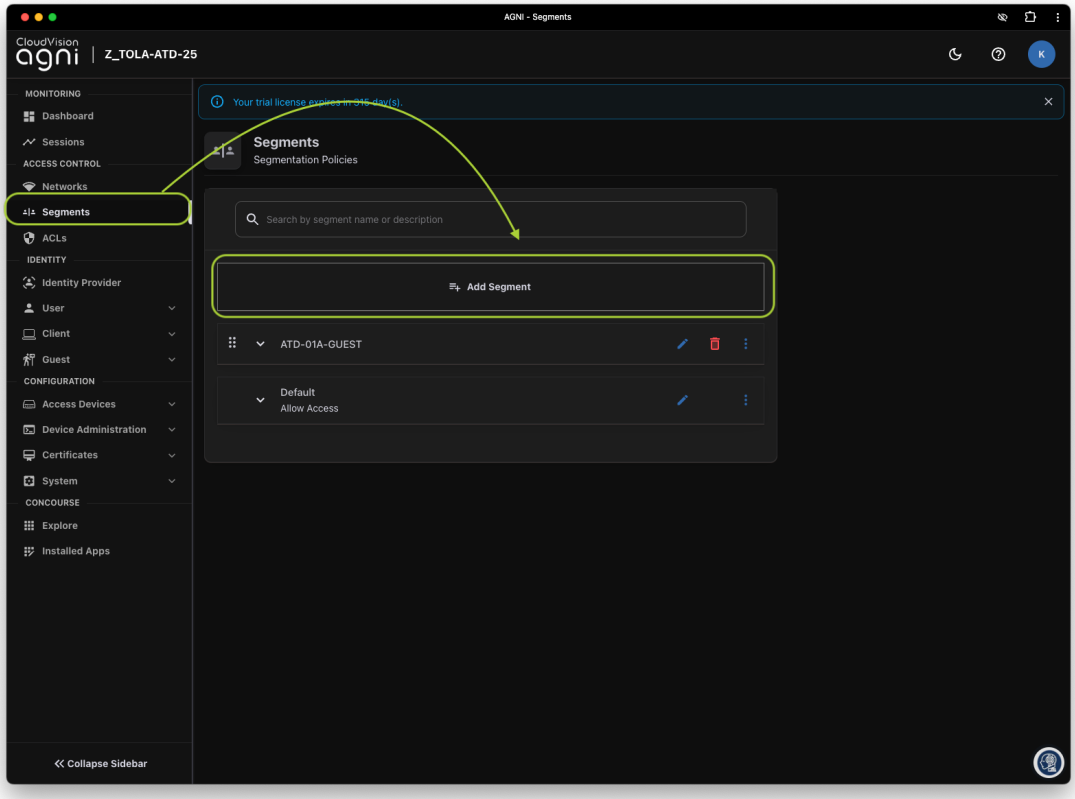
Network Settings

Field	Student 1	Student 2
Name	`ATD-##A-EAP`	`ATD-##B-EAP`
Connection Type	Wireless	Wireless
SSID	`ATD-##A-EAP`	`ATD-##B-EAP`
Authentication Type	802.1X EAP	802.1X EAP
EAP Methods	EAP-TLS	EAP-TLS



3. Select `Add Network` at the bottom of the screen.

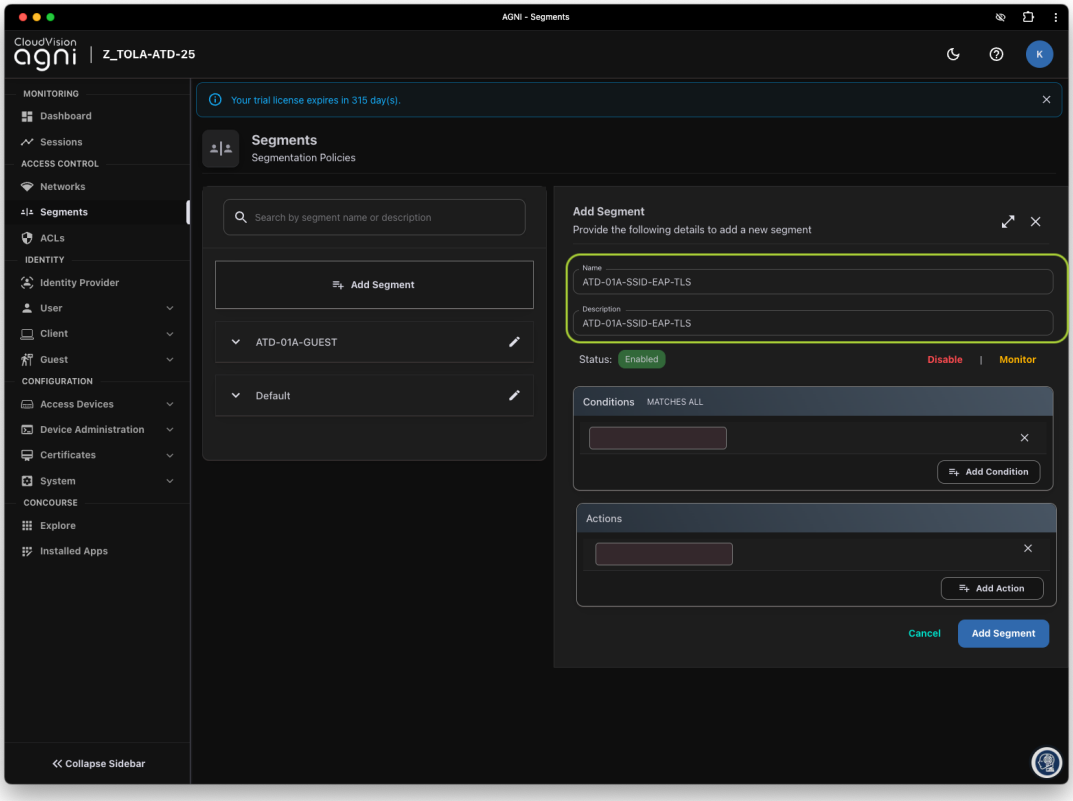
4. Next, select `Segments` and then `+ Add Segment`



5. Configure the segment with the following settings

Network Settings

Field	Student 1	Student 2
Name	`ATD-##A-EAP`	`ATD-##B-EAP`
Description	`ATD-##A-EAP`	`ATD-##B-EAP`



6. Next, let's add two conditions to match the network we've defined (tied to the SSID) and the authentication type

Conditions

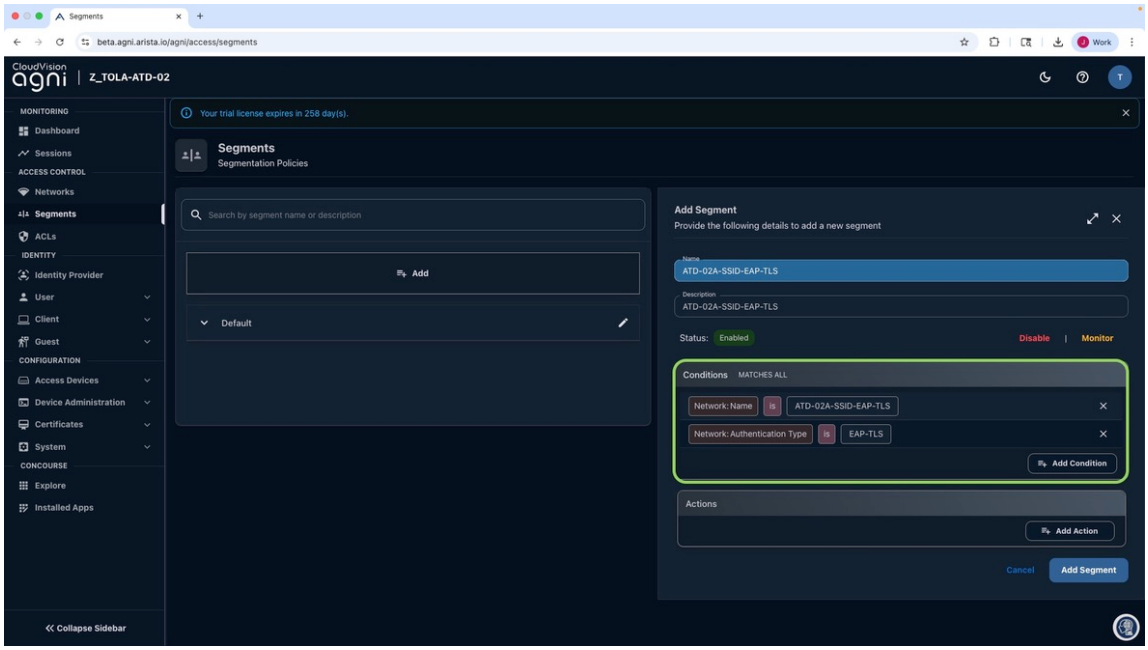
Conditions for segments must MATCH ALL conditions line by line.

a. Select `Network`, `Name`, `is`, `ATD-##X-EAP` from the dropdown lists. Choose your `A` or `B`

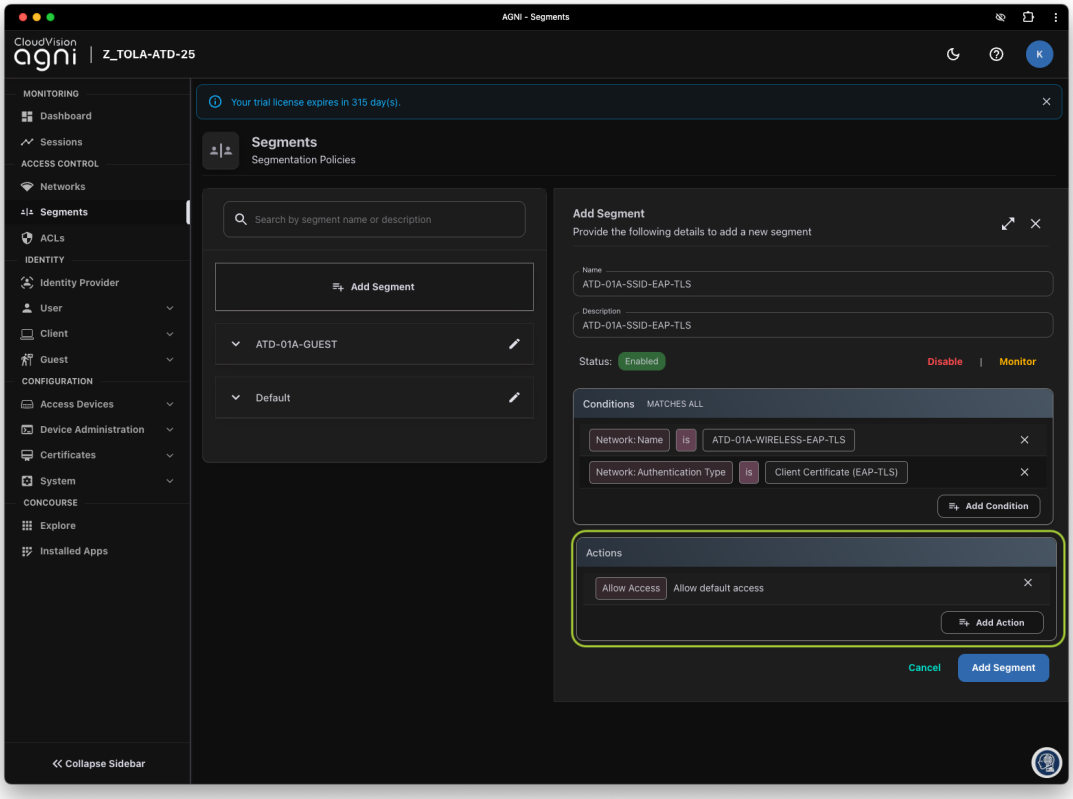
policy accordingly.

b. Select, `Network`, `Authentication Type`, `is`, `Client Certificate (EAP-TLS)` from the drop down lists.

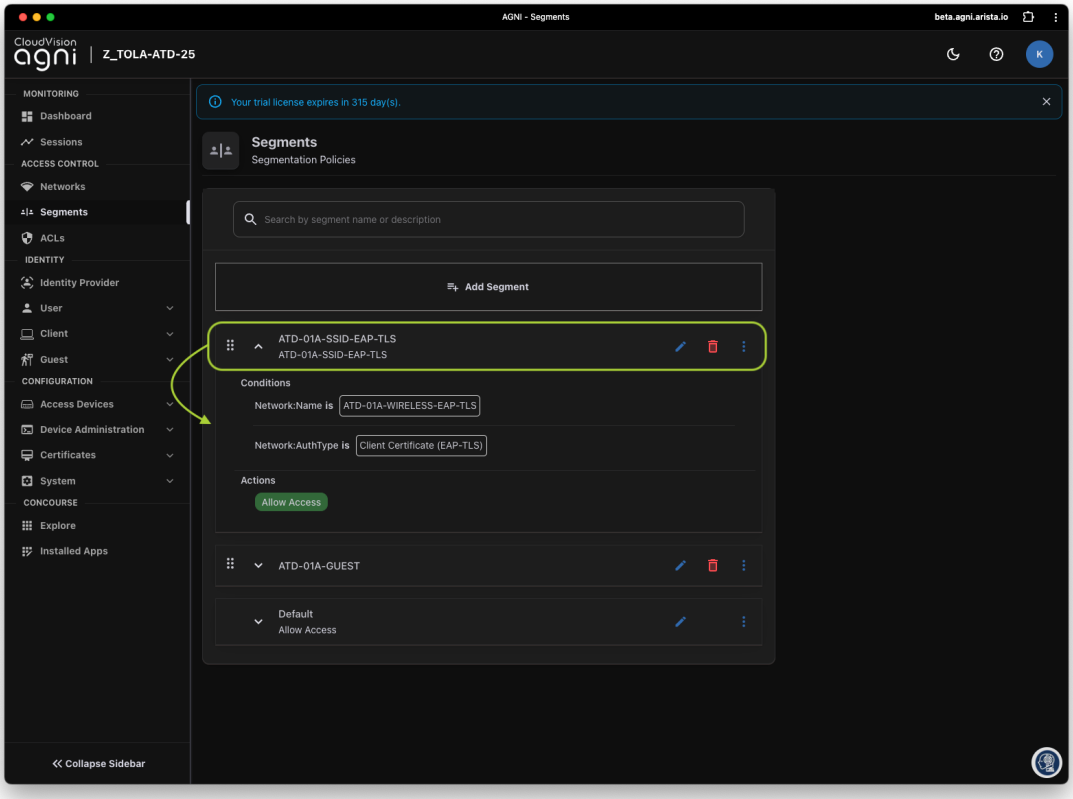
c. Your Conditions should now look like this.



7. Under `Actions` select `Add Action` and select `Allow Access`

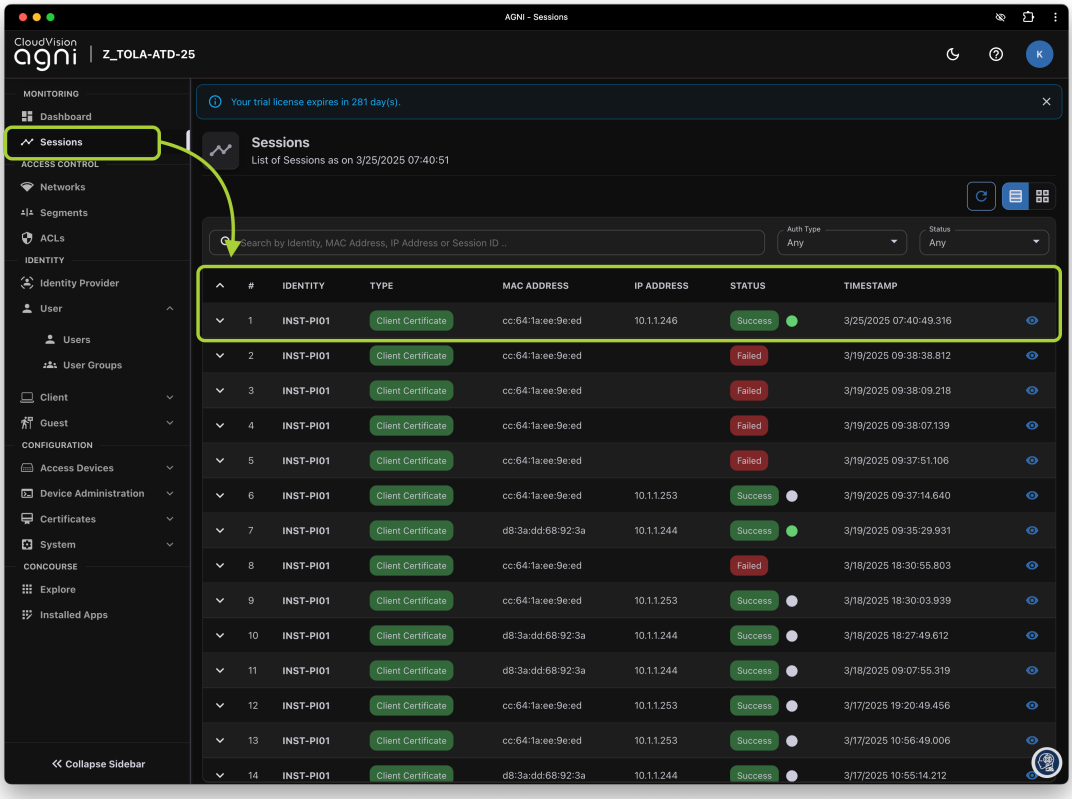


8. Finally, select `Add Segment` at the bottom of the page.
9. You should now be able to expand and review your segment.

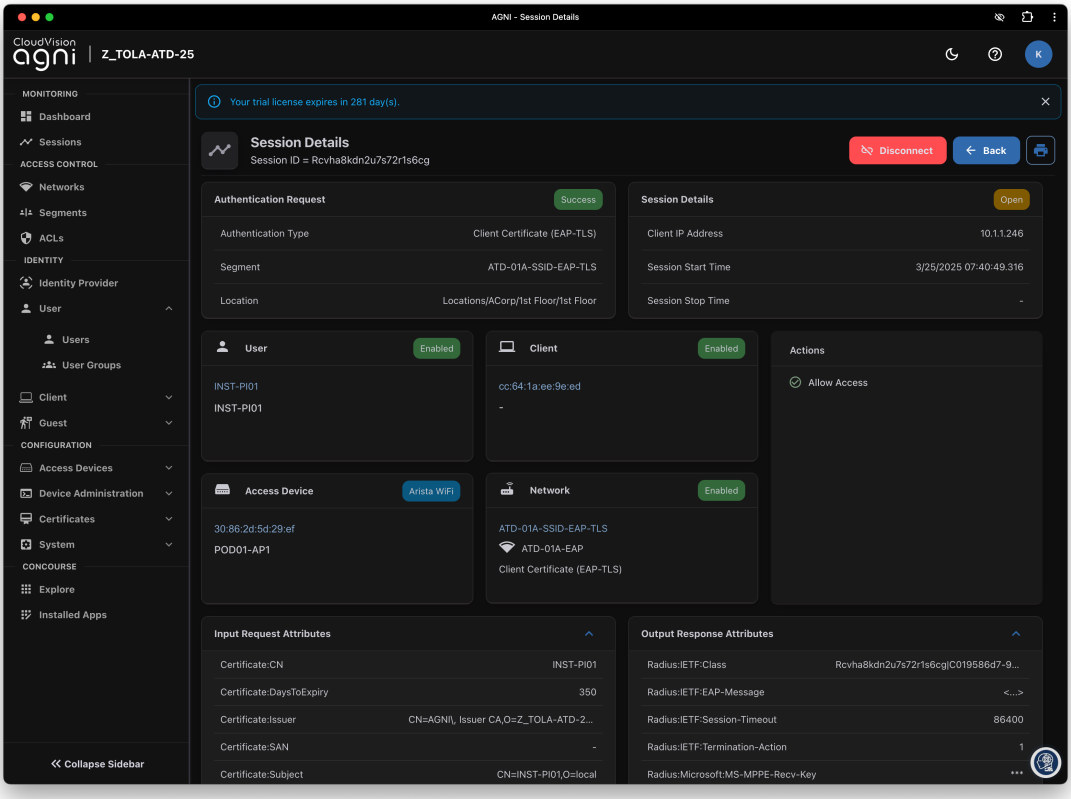


10. Next, select `Sessions` to see if your ATD Raspberry Pi has a connection via the Wireless connection.

> **Client Connectivity**
>
> The Client Certificate has already been applied to the Raspberry Pi and is configured to connect to the SSID ATD-##A-EAP.



11. Select the session and explore the information we learn about the client. We're going to come back to this in more detail later.



12. If you don’t see any new sessions within 2 minutes in AGNI, power cycle the Raspberry Pi.

Additional References

- Configuring RadSec
- EAP-TLS with MacBook

🎉 CONGRATS! You have completed this lab! 🎉

LET'S GO TO THE NEXT LAB!

2026-06-10